

Ion flow modeling of insect odorant receptors *via* Computational Electrophysiology

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ABSTRACT

Insects detect odors through odorant-activated ion channel complexes that are composed of an odorant receptor (OR) and an odorant receptor co-receptor (Orco), which are thought to exist in a 1:3 OR/Orco ratio in vivo^{1,2}. Recently published cryo-EM structures of OR/Orco complexes from mosquitoes with bound known agonists (AgamOR28 from *Anopheles gambiae* with 2,4,5-trimethylthiazole, and AaegOR10 from *Aedes aegypti* with o-cresol^{1,3-4}), showed a displacement of the transmembrane helix 7b (TM7b) of the ORs which allows ions to flow through the ion channel. To elucidate the mechanism of the ion flow, we used Computational Electrophysiology (CompEL)⁵ protocol using Amber force fields⁶ and the molecular dynamic (MD) code Gromacs⁷. Ion flow was observed for the apo structure, see Fig. 1, likewise for agonist bound structures. Comparing the different systems revealed key-residues and a side entrance to the pore.

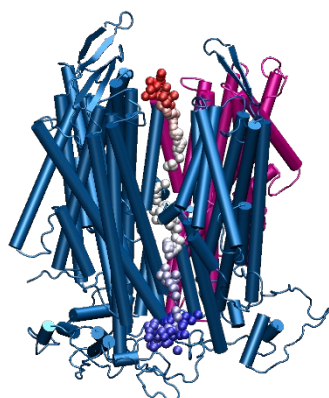


Fig. 1 A side-on view of the receptor system, AaegOR10. The OR subunit is colored magenta while Orco is dark cerulean. The path of a single potassium ion through the pore is shown, colored by evolution of time, red for early on, white for halftime and blue for late evolution.

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